Name:S.Shambhavi H.No:2303A51832 Batch:26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		**Assignment Type: Lab**	**Academic Year:**2025-2026
Course Coordinator Name	Dr. Rishabh Mittal		
Instructor(s) Name	Mr. S Naresh Kumar Ms. B. Swathi Dr. Sasanko Shekhar Gantayat Mr. Md Sallauddin Dr. Mathivanan Mr. Y Srikanth Ms. N Shilpa Dr. Rishabh Mittal (Coordinator) Dr. R. Prashant Kumar Mr. Ankushavali MD Mr. B Viswanath Ms. Sujitha Reddy Ms. A. Anitha Ms. M.Madhuri Ms. Katherashala Swetha Ms. Velpula sumalatha Mr. Bingi Raju		
Course Code	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/II	**Regulation**	R23
Date and Day of Assignment	**Week6 – Wednesday**	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicable to Batches**	All batches
Assignment Number:11.3(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*Expected Time to complete*
1	**Lab 11: Data Structures with AI Implementing Fundamental Data Structures using AI Assistance** **Lab Objectives:**	Week6 - Wednesday

By the end of this lab, students will be able to:

- Design and implement fundamental data structures in Python using AI assistance.
- Effectively prompt AI tools (e.g., GitHub Copilot) for code generation, optimization, and documentation.
- Understand and compare core data structures: Arrays, Linked Lists, Stacks, Queues, Priority Queues, Trees, and Graphs.
- Improve code readability, efficiency, and maintainability using AI-generated suggestions.

Learning Outcomes

After completing this lab, students will be able to:

- Apply appropriate data structures to solve real-world problems.
- Analyze time and space complexity of different data structure operations.
- Use AI tools responsibly to assist (not replace) logical thinking and problem-solving.
- Validate, test, and refine AI-generated code.

Task 1: Smart Contact Manager (Arrays & Linked Lists)

Scenario

SR University's student club requires a simple **Contact Manager Application** to store members' names and phone numbers. The system should support efficient addition, searching, and deletion of contacts.

Tasks

1. Implement the contact manager using **arrays (lists)**.
2. Implement the same functionality using a **linked list** for dynamic memory allocation.
3. Implement the following operations in both approaches:
    - Add a contact
    - Search for a contact
    - Delete a contact
4. Use **GitHub Copilot** to assist in generating search and delete methods.
5. Compare **array vs. linked list** approaches with respect to:
    - Insertion efficiency
    - Deletion efficiency

Expected Outcome

- Two working implementations (array-based and linked-list-based).
- A brief comparison explaining performance differences.

Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ArrayContactManager

```python
class ArrayContactManager:
    def __init__(self):
        self.contacts = []
    def add_contact(self, name, phone):
        self.contacts.append({'name': name, 'phone': phone})
    def search_contact(self, name):
        for contact in self.contacts:
            if contact['name'] == name:
                return contact['phone']
        return None
    def delete_contact(self, name):
        for i, contact in enumerate(self.contacts):
            if contact['name'] == name:
                del self.contacts[i]
                return True
        return False

class Node:
    def __init__(self, name, phone):
        self.name = name
        self.phone = phone
        self.next = None

class LinkedListContactManager:
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\shash\Downloads> conda activate Shashidhar
PS C:\Users\shash\Downloads>  & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions
\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '63778' '--' 'c:\Users\shash\Downloads\AAC 11.4.py
'

Array search Alice: 123
Array delete Bob: True
Array search Bob after delete: None
Linked search Charlie: 789
Linked delete Dave: True
Linked search Dave after delete: None
PS C:\Users\shash\Downloads>
```

powersh... | Python Deb... | 1 Hidden T...

CHAT

CONTACT MANAGER IMPLEMENTATION WITH ARRAYS AND LINKED LI...

- o Add a contact
- o Search for a contact
- o Delete a contact

4. Use GitHub Copilot to assist in generating search and delete methods.
5. Compare array vs. linked list approaches with respect to:
- o Insertion efficiency
- o Deletion efficiency

Expected Outcome

- Two working implementations (array-based and linked-list-based).
- A brief comparison explaining performance differences.

give code in python without comments

✓ pwd

✓ Created array_contact_manager.py

✓ Created linked_list_contact_manager.py

✓ Created comparison.txt

I've created three files in your current directory (shash) with the implementations:

- `array_contact_manager.py`: Array-based contact manager
- `linked_list_contact_manager.py`: Linked list-based contact manager
- `comparison.txt`: Brief performance comparison

The code is provided without comments as requested. You can run these files directly in Python to test the functionality. For example, you could create a simple test script to instantiate the classes and call the methods.

AAC 11.4.py

Describe what to build next

Grok Code Fast 1

Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ArrayContactManager

```python
class Node:
    def __init__(self, name, phone):
        self.phone = phone
        self.next = None

class LinkedListContactManager:
    def __init__(self):
        self.head = None
    def add_contact(self, name, phone):
        new_node = Node(name, phone)
        if not self.head:
            self.head = new_node
        else:
            current = self.head
            while current.next:
                current = current.next
            current.next = new_node
    def search_contact(self, name):
        current = self.head
        while current:
            if current.name == name:
                return current.phone
            current = current.next
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\shash\Downloads> conda activate Shashidhar
PS C:\Users\shash\Downloads>  & 'c:\Users\shash\anaconda3\envs\Shashidh
\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '
'

Array search Alice: 123
Array delete Bob: True
Array search Bob after delete: None
Linked search Charlie: 789
Linked delete Dave: True
Linked search Dave after delete: None
PS C:\Users\shash\Downloads>
```

Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ArrayContactManager

```python
class LinkedListContactManager:
    def search_contact(self, name):
                return current.phone
            current = current.next
        return None
    def delete_contact(self, name):
        if not self.head:
            return False
        if self.head.name == name:
            self.head = self.head.next
            return True
        current = self.head
        while current.next:
            if current.next.name == name:
                current.next = current.next.next
                return True
            current = current.next
        return False

array_cm = ArrayContactManager()
array_cm.add_contact("Alice", "123")
array_cm.add_contact("Bob", "456")
print("Array search Alice:", array_cm.search_contact("Alice"))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads> conda activate Shashidhar
PS C:\Users\shash\Downloads>  & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscode\extensions
\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '63778' '--' 'c:\Users\shash\Downloads\AAC 11.4.py
'
Array search Alice: 123
Array delete Bob: True
Array search Bob after delete: None
Linked search Charlie: 789
Linked delete Dave: True
Linked search Dave after delete: None
PS C:\Users\shash\Downloads>
```

Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ArrayContactManager

```python
class LinkedListContactManager:
    def delete_contact(self, name):
            current = current.next
        return False

array_cm = ArrayContactManager()
array_cm.add_contact("Alice", "123")
array_cm.add_contact("Bob", "456")
print("Array search Alice:", array_cm.search_contact("Alice"))
print("Array delete Bob:", array_cm.delete_contact("Bob"))
print("Array search Bob after delete:", array_cm.search_contact("Bob"))

linked_cm = LinkedListContactManager()
linked_cm.add_contact("Charlie", "789")
linked_cm.add_contact("Dave", "101")
print("Linked search Charlie:", linked_cm.search_contact("Charlie"))
print("Linked delete Dave:", linked_cm.delete_contact("Dave"))
print("Linked search Dave after delete:", linked_cm.search_contact("Dave"))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads> conda activate Shashidhar
PS C:\Users\shash\Downloads>  & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\.vscod
\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '63778' '--' 'c:\Users\shash\Downloads
'
Array search Alice: 123
Array delete Bob: True
Array search Bob after delete: None
Linked search Charlie: 789
Linked delete Dave: True
Linked search Dave after delete: None
PS C:\Users\shash\Downloads>
```

Task 2: Library Book Search System (Queues & Priority Queues)

Scenario

The SRU Library manages book borrow requests. Students and faculty submit requests, but **faculty requests must be prioritized** over student requests.

Tasks

1. Implement a **Queue (FIFO)** to manage book requests.
2. Extend the system to a **Priority Queue**, prioritizing faculty requests.
3. Use **GitHub Copilot** to assist in generating:
    - enqueue() method
    - dequeue() method
4. Test the system with a mix of student and faculty requests.

Expected Outcome

- Working queue and priority queue implementations.
- Correct prioritization of faculty requests.

Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
from collections import deque
import heapq
class BookRequest:
    def __init__(self, requester_name, book_title, role):
        self.requester_name = requester_name
        self.book_title = book_title
        self.role = role
    def __repr__(self):
        return f"{self.role.upper()} | {self.requester_name} -> '{self.book_title}'"
class LibraryQueue:
    def __init__(self):
        self._queue = deque()
    def enqueue(self, request):
        self._queue.append(request)
        print(f"[QUEUE] Added: {request}")
    def dequeue(self):
        if not self._queue:
            print("[QUEUE] No requests in queue.")
            return None
        request = self._queue.popleft()
        print(f"[QUEUE] Processing: {request}")
        return request
    def is_empty(self):
        return len(self._queue) == 0
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '59531' '--' 'c
:\Users\shash\Downloads\AAC 11.4.py'

--- Dequeuing all requests ---
[PRIORITY QUEUE] Processing: FACULTY | Dr. Smith -> 'Algorithms'
[PRIORITY QUEUE] Processing: FACULTY | Dr. Lee -> 'Database Systems'
[PRIORITY QUEUE] Processing: STUDENT | Alice -> 'Data Structures'
[PRIORITY QUEUE] Processing: STUDENT | Bob -> 'Operating Systems'
[PRIORITY QUEUE] Processing: STUDENT | Carol -> 'Computer Networks'
```

CHAT

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION

Library Book Search System (Queues & Priority Queues)
Scenario
The SRU Library manages book borrow requests. Students and faculty submit requests, but faculty requests must be prioritized over student requests.
Tasks
1. Implement a Queue (FIFO) to manage book requests.
2. Extend the system to a Priority Queue, prioritizing faculty requests.
3. Use GitHub Copilot to assist in generating:
o enqueue() method
o dequeue() method
4. Test the system with a mix of student and faculty requests.
Expected Outcome
• Working queue and priority queue implementations.
• Correct prioritization of faculty requests.
give code in python without comments just paste in the file

✓ Optimizing tool selection...
✓ Reviewed and created AAC 11.4.py
✓ Created AAC 11.4.py
✓ Configured a Python environment
✓ Ran `pylanceRunCodeSnippet` - pylance mcp server (MCP Server)
✓ Reviewed code and provided a brief summary

AAC 11.4.py
Describe what to build next
Claude Sonnet 4.6

Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
class LibraryQueue:
    def is_empty(self):
        return len(self._queue) == 0
    def display(self):
        if not self._queue:
            print("[QUEUE] Queue is empty.")
        else:
            print("[QUEUE] Current requests:")
            for req in self._queue:
                print(f"  {req}")
class PriorityLibraryQueue:
    PRIORITY_MAP = {"faculty": 0, "student": 1}

    def __init__(self):
        self._heap = []
        self._counter = 0
    def enqueue(self, request):
        priority = self.PRIORITY_MAP.get(request.role.lower(), 2)
        heapq.heappush(self._heap, (priority, self._counter, request))
        self._counter += 1
        print(f"[PRIORITY QUEUE] Added: {request}")
    def dequeue(self):
        if not self._heap:
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '59531' '--' 'c
:\Users\shash\Downloads\AAC 11.4.py'

--- Dequeuing all requests ---
[PRIORITY QUEUE] Processing: FACULTY | Dr. Smith -> 'Algorithms'
[PRIORITY QUEUE] Processing: FACULTY | Dr. Lee -> 'Database Systems'
[PRIORITY QUEUE] Processing: STUDENT | Alice -> 'Data Structures'
[PRIORITY QUEUE] Processing: STUDENT | Bob -> 'Operating Systems'
[PRIORITY QUEUE] Processing: STUDENT | Carol -> 'Computer Networks'
```

CHAT

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION

✓ Optimizing tool selection...
✓ Reviewed and created AAC 11.4.py
✓ Created AAC 11.4.py
✓ Configured a Python environment
✓ Ran `pylanceRunCodeSnippet` - pylance mcp server (MCP Server)
✓ Reviewed code and provided a brief summary

Code written to [AAC 11.4.py](c:\Users\shash\Downloads\AAC 11.4.py) and verified working. Here's what the output shows:

Regular Queue (FIFO): Processes in insertion order — Alice, Dr. Smith, Bob, Dr. Lee, Carol.

Priority Queue: Automatically reorders so faculty always come first — Dr. Smith, Dr. Lee (faculty), then Alice, Bob, Carol (students). Within the same role, arrival order (FIFO) is preserved.

Key classes:

- `BookRequest` — holds requester name, book title, and role
- `LibraryQueue` — FIFO using `collections.deque` with `enqueue()`/`dequeue()`
- `PriorityLibraryQueue` — min-heap via `heapq` assigning priority `0` to faculty and `1` to students

Claude Sonnet 4.6 • 1x

AAC 11.4.py
Describe what to build next
Claude Sonnet 4.6

Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
class PriorityLibraryQueue:
    def dequeue(self):
            print("[PRIORITY QUEUE] No requests in queue.")
            return None
        _, _, request = heapq.heappop(self._heap)
        print(f"[PRIORITY QUEUE] Processing: {request}")
        return request
    def is_empty(self):
        return len(self._heap) == 0
    def display(self):
        if not self._heap:
            print("[PRIORITY QUEUE] Queue is empty.")
        else:
            sorted_requests = sorted(self._heap, key=lambda x: (x[0], x[1]))
            print("[PRIORITY QUEUE] Current requests (by priority):")
            for _, _, req in sorted_requests:
                print(f"  {req}")
print("=" * 55)
print("         REGULAR QUEUE (FIFO) TEST")
print("=" * 55)
library_queue = LibraryQueue()
library_queue.enqueue(BookRequest("Alice", "Data Structures", "student"))
library_queue.enqueue(BookRequest("Dr. Smith", "Algorithms", "faculty"))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\S
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\laun
:\Users\shash\Downloads\AAC 11.4.py'

--- Dequeuing all requests ---
[PRIORITY QUEUE] Processing: FACULTY | Dr. Smith -> 'Algorithms'
[PRIORITY QUEUE] Processing: FACULTY | Dr. Lee -> 'Database Systems'
[PRIORITY QUEUE] Processing: STUDENT | Alice -> 'Data Structures'
[PRIORITY QUEUE] Processing: STUDENT | Bob -> 'Operating Systems'
[PRIORITY QUEUE] Processing: STUDENT | Carol -> 'Computer Networks'
```

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
library_queue.enqueue(BookRequest("Dr. Smith", "Algorithms", "faculty"))
library_queue.enqueue(BookRequest("Bob", "Operating Systems", "student"))
library_queue.enqueue(BookRequest("Dr. Lee", "Database Systems", "faculty"))
library_queue.enqueue(BookRequest("Carol", "Computer Networks", "student"))
print()
library_queue.display()
print()
print("--- Dequeuing all requests ---")
while not library_queue.is_empty():
    library_queue.dequeue()
print()
print("=" * 55)
print("       PRIORITY QUEUE TEST (Faculty First)")
print("=" * 55)
priority_queue = PriorityLibraryQueue()
priority_queue.enqueue(BookRequest("Alice", "Data Structures", "student"))
priority_queue.enqueue(BookRequest("Dr. Smith", "Algorithms", "faculty"))
priority_queue.enqueue(BookRequest("Bob", "Operating Systems", "student"))
priority_queue.enqueue(BookRequest("Dr. Lee", "Database Systems", "faculty"))
priority_queue.enqueue(BookRequest("Carol", "Computer Networks", "student"))
print()
priority_queue.display()
print()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugp
:\Users\shash\Downloads\AAC 11.4.py'

--- Dequeuing all requests ---
[PRIORITY QUEUE] Processing: FACULTY | Dr. Smith -> 'Algorithms'
[PRIORITY QUEUE] Processing: FACULTY | Dr. Lee -> 'Database Systems'
[PRIORITY QUEUE] Processing: STUDENT | Alice -> 'Data Structures'
[PRIORITY QUEUE] Processing: STUDENT | Bob -> 'Operating Systems'
[PRIORITY QUEUE] Processing: STUDENT | Carol -> 'Computer Networks'
```

Welcome AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
print("=" * 55)
print("       PRIORITY QUEUE TEST (Faculty First)")
print("=" * 55)
priority_queue = PriorityLibraryQueue()
priority_queue.enqueue(BookRequest("Alice", "Data Structures", "student"))
priority_queue.enqueue(BookRequest("Dr. Smith", "Algorithms", "faculty"))
priority_queue.enqueue(BookRequest("Bob", "Operating Systems", "student"))
priority_queue.enqueue(BookRequest("Dr. Lee", "Database Systems", "faculty"))
priority_queue.enqueue(BookRequest("Carol", "Computer Networks", "student"))
print()
priority_queue.display()
print()
print("--- Dequeuing all requests ---")
while not priority_queue.is_empty():
    priority_queue.dequeue()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anacond
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\deb
:\Users\shash\Downloads\AAC 11.4.py'

--- Dequeuing all requests ---
[PRIORITY QUEUE] Processing: FACULTY | Dr. Smith -> 'Algorithms'
[PRIORITY QUEUE] Processing: FACULTY | Dr. Lee -> 'Database Systems'
[PRIORITY QUEUE] Processing: STUDENT | Alice -> 'Data Structures'
[PRIORITY QUEUE] Processing: STUDENT | Bob -> 'Operating Systems'
[PRIORITY QUEUE] Processing: STUDENT | Carol -> 'Computer Networks'
PS C:\Users\shash\Downloads>
```

Task 3: Emergency Help Desk (Stack Implementation)

Scenario

SR University's IT Help Desk receives technical support tickets from students and staff. While tickets are received sequentially, issue escalation follows a **Last-In, First-Out (LIFO)** approach.

Tasks

1. Implement a **Stack** to manage support tickets.
2. Provide the following operations:
   - push(ticket)
   - pop()
   - peek()
3. Simulate at least **five tickets** being raised and resolved.
4. Use **GitHub Copilot** to suggest additional stack operations such as:
   - Checking whether the stack is empty
   - Checking whether the stack is full (if applicable)

Expected Outcome

- Functional stack-based ticket management system.
- Clear demonstration of LIFO behavior.

```python
class SupportTicket:
    def __init__(self, ticket_id, requester, issue, priority="normal"):
        self.ticket_id = ticket_id
        self.requester = requester
        self.issue = issue
        self.priority = priority

    def __repr__(self):
        return f"[#{self.ticket_id}] {self.requester} | {self.issue} ({self.priority.upper()})"


class HelpDeskStack:
    MAX_SIZE = 10

    def __init__(self):
        self._stack = []

    def push(self, ticket):
        if self.is_full():
            print(f"[STACK FULL] Cannot add: {ticket}")
            return
        self._stack.append(ticket)
        print(f"[PUSHED] {ticket}")
```

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '51957' '--' 'c
:\Users\shash\Downloads\AAC 11.4.py'
      SR UNIVERSITY IT HELP DESK - TICKET SYSTEM
==========================================================

--- Raising Support Tickets ---
[PUSHED] [#101] Alice (Student) | Cannot access Wi-Fi (NORMAL)
[PUSHED] [#102] Bob (Staff) | Projector not working in Lab3 (HIGH)
[PUSHED] [#103] Carol (Student) | Email login failure (NORMAL)
[PUSHED] [#104] Dr. Smith (Staff) | VPN disconnecting repeatedly (CRITICAL)
```

CHAT

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION

o push(ticket)
o pop()
o peek()
3. Simulate at least five tickets being raised and resolved.
4. Use GitHub Copilot to suggest additional stack operations such as:
o Checking whether the stack is empty
o Checking whether the stack is full (if applicable)
Expected Outcome
• Functional stack-based ticket management system.
• Clear demonstration of LIFO behavior.
give code in python without comments

✓ Created AAC 11.5.py
✓ Created AAC 11.5.py
✓ Tried to run command but was blocked
✓ Ran `pylance#runCodeSnippet` – pylance mcp server (MCP Server)
✓ Summarized results for the user

Code saved to [AAC 11.5.py](c:\Users\shash\Downloads\AAC 11.5.py) and verified working. Here's what it demonstrates:

Classes:

- `SupportTicket` — holds ticket ID, requester, issue description, and priority
- `HelpDeskStack` — list-based stack with `MAX_SIZE = 10`

AAC 11.4.py
Describe what to build next
Claude Sonnet 4.6

```python
    print(f"[PUSHED] {ticket}")

    def pop(self):
        if self.is_empty():
            print("[STACK EMPTY] No tickets to resolve.")
            return None
        ticket = self._stack.pop()
        print(f"[RESOLVED] {ticket}")
        return ticket

    def peek(self):
        if self.is_empty():
            print("[STACK EMPTY] No tickets in stack.")
            return None
        ticket = self._stack[-1]
        print(f"[PEEK] Next to resolve: {ticket}")
        return ticket

    def is_empty(self):
        return len(self._stack) == 0

    def is_full(self):
```

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '51957' '--' 'c
:\Users\shash\Downloads\AAC 11.4.py'
[PUSHED] [#105] Eve (Student) | Printer offline in Library (NORMAL)

[STACK] Current tickets (top = most recent, total=5):
  [#105] Eve (Student) | Printer offline in Library (NORMAL)
  [#104] Dr. Smith (Staff) | VPN disconnecting repeatedly (CRITICAL)
  [#103] Carol (Student) | Email login failure (NORMAL)
  [#102] Bob (Staff) | Projector not working in Lab3 (HIGH)
  [#101] Alice (Student) | Cannot access Wi-Fi (NORMAL)
```

CHAT

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION

✓ Created AAC 11.5.py
✓ Tried to run command but was blocked
✓ Ran `pylance#runCodeSnippet` – pylance mcp server (MCP Server)
✓ Summarized results for the user

Code saved to [AAC 11.5.py](c:\Users\shash\Downloads\AAC 11.5.py) and verified working. Here's what it demonstrates:

Classes:

- `SupportTicket` — holds ticket ID, requester, issue description, and priority
- `HelpDeskStack` — list-based stack with `MAX_SIZE = 10`

Operations implemented:

Method	Description
`push(ticket)`	Adds ticket to top of stack
`pop()`	Removes and resolves ticket from top (LIFO)
`peek()`	Views top ticket without removing it
`is_empty()`	Returns `True` if no tickets
`is_full()`	Returns `True` if at `MAX_SIZE`
`size()`	Returns current ticket count
`display()`	Shows all tickets top-to-bottom

AAC 11.4.py
Describe what to build next
Claude Sonnet 4.6

```python
    def is_empty(self):
        return len(self._stack) == 0

    def is_full(self):
        return len(self._stack) >= self.MAX_SIZE

    def size(self):
        return len(self._stack)

    def display(self):
        if self.is_empty():
            print("[STACK] Stack is empty.")
        else:
            print(f"[STACK] Current tickets (top = most recent, total={self.size()}):")
            for ticket in reversed(self._stack):
                print(f"  {ticket}")


print("-" * 60)
print("      SR UNIVERSITY IT HELP DESK - TICKET SYSTEM")
print("-" * 60)
```

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '51957' '--' 'c
:\Users\shash\Downloads\AAC 11.4.py'
  [#101] Alice (Student) | Cannot access Wi-Fi (NORMAL)

[INFO] Stack empty? False
[INFO] Stack full?  False
[INFO] Total tickets in stack: 5

--- Peeking at top ticket ---
[PEEK] Next to resolve: [#105] Eve (Student) | Printer offline in Library (NORMAL)
```

CHAT

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION

✓ Summarized results for the user

Code saved to [AAC 11.5.py](c:\Users\shash\Downloads\AAC 11.5.py) and verified working. Here's what it demonstrates:

Classes:

- `SupportTicket` — holds ticket ID, requester, issue description, and priority
- `HelpDeskStack` — list-based stack with `MAX_SIZE = 10`

Operations implemented:

Method	Description
`push(ticket)`	Adds ticket to top of stack
`pop()`	Removes and resolves ticket from top (LIFO)
`peek()`	Views top ticket without removing it
`is_empty()`	Returns `True` if no tickets
`is_full()`	Returns `True` if at `MAX_SIZE`
`size()`	Returns current ticket count
`display()`	Shows all tickets top-to-bottom

LIFO behavior confirmed: Tickets were pushed in order 101→105, and resolved in reverse order 105→101. The final `pop()` on an empty stack is handled gracefully with a message instead of crashing.

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AAC 11.4.py
Describe what to build next
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Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
print("=" * 60)
print("       SR UNIVERSITY IT HELP DESK - TICKET SYSTEM")
print("=" * 60)

help_desk = HelpDeskStack()

print("\n--- Raising Support Tickets ---")
help_desk.push(SupportTicket(101, "Alice (Student)",   "Cannot access Wi-Fi",          "normal"))
help_desk.push(SupportTicket(102, "Bob (Staff)",       "Projector not working in Lab3", "high"))
help_desk.push(SupportTicket(103, "Carol (Student)",   "Email login failure",          "normal"))
help_desk.push(SupportTicket(104, "Dr. Smith (Staff)", "VPN disconnecting repeatedly",  "critical"))
help_desk.push(SupportTicket(105, "Eve (Student)",     "Printer offline in Library",   "normal"))

print()
help_desk.display()

print(f"\n[INFO] Stack empty? {help_desk.is_empty()}")
print(f"[INFO] Stack full?  {help_desk.is_full()}")
print(f"[INFO] Total tickets in stack: {help_desk.size()}")

print("\n--- Peeking at top ticket ---")
help_desk.peek()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '51957' '--'
:\Users\shash\Downloads\AAC 11.4.py'
--- Peeking at top ticket ---
[PEEK] Next to resolve: [#105] Eve (Student) | Printer offline in Library (NORMAL)

--- Resolving Tickets (LIFO Order) ---
[RESOLVED] [#105] Eve (Student) | Printer offline in Library (NORMAL)
[RESOLVED] [#104] Dr. Smith (Staff) | VPN disconnecting repeatedly (CRITICAL)
[RESOLVED] [#103] Carol (Student) | Email login failure (NORMAL)
[RESOLVED] [#102] Bob (Staff) | Projector not working in Lab3 (HIGH)
```

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
print()
help_desk.display()

print(f"\n[INFO] Stack empty? {help_desk.is_empty()}")
print(f"[INFO] Stack full?  {help_desk.is_full()}")
print(f"[INFO] Total tickets in stack: {help_desk.size()}")

print("\n--- Peeking at top ticket ---")
help_desk.peek()

print("\n--- Resolving Tickets (LIFO Order) ---")
while not help_desk.is_empty():
    help_desk.pop()

print()
print(f"[INFO] Stack empty after resolving all? {help_desk.is_empty()}")
help_desk.pop()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\an
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\lib
:\Users\shash\Downloads\AAC 11.4.py'
[RESOLVED] [#104] Dr. Smith (Staff) | VPN disconnecting repeatedly (CRITICAL)
[RESOLVED] [#103] Carol (Student) | Email login failure (NORMAL)
[RESOLVED] [#102] Bob (Staff) | Projector not working in Lab3 (HIGH)
[RESOLVED] [#101] Alice (Student) | Cannot access Wi-Fi (NORMAL)

[INFO] Stack empty after resolving all? True
[STACK EMPTY] No tickets to resolve.
PS C:\Users\shash\Downloads>
```

Task 4: Hash Table

Objective

To implement a **Hash Table** and understand collision handling.

Task Description

Use AI to generate a hash table with:

- Insert
- Search
- Delete

Starter Code

class HashTable:

pass

Expected Outcome

- Collision handling using chaining
- Well-commented methods

Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
class HashTable:
    def __init__(self, capacity=16):
        self.capacity = capacity
        self.size = 0
        self.buckets = [[] for _ in range(self.capacity)]

    def _hash(self, key):
        return hash(key) % self.capacity

    def insert(self, key, value):
        index = self._hash(key)
        bucket = self.buckets[index]
        for i, (k, v) in enumerate(bucket):
            if k == key:
                bucket[i] = (key, value)
                return
        bucket.append((key, value))
        self.size += 1

    def search(self, key):
        index = self._hash(key)
        bucket = self.buckets[index]
        for k, v in bucket:
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60955' '--' 'c
:\Users\shash\Downloads\AAC 11.4.py'

[HASH TABLE] capacity=7, entries=8, load_factor=1.14
  Bucket[ 0]: (frank: frank@sru.edu)
  Bucket[ 2]: (alice: alice.new@sru.edu) -> (carol: carol@sru.edu) -> (eve: eve@sru.edu)
  Bucket[ 3]: (dave: dave@sru.edu) -> (heidi: heidi@sru.edu)
  Bucket[ 4]: (judy: judy@sru.edu)
  Bucket[ 6]: (ivan: ivan@sru.edu)
```

CHAT

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION

Hash Table
Objective
To implement a Hash Table and understand collision handling.
Task Description
Use AI to generate a hash table with:
• Insert
• Search
• Delete
Starter Code
class HashTable:
pass
Expected Outcome
• Collision handling using chaining
• Well-commented methods
give code in python without comments'

✓ Created AAC 11.6.py
✓ Created AAC 11.6.py
✓ Tried to run command but was blocked
✓ Ran `pylanceRunCodeSnippet` – pylance mcp server (MCP Server)
✓ Reviewed output for the user

Code saved to [AAC 11.6.py](c:\Users\shash\Downloads\AAC 11.6.py) and verified working. Here's a summary:

AAC 11.4.py
Describe what to build next
Claude Sonnet 4.6

Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
class HashTable:
    def search(self, key):
        bucket = self.buckets[index]
        for k, v in bucket:
            if k == key:
                return v
        return None

    def delete(self, key):
        index = self._hash(key)
        bucket = self.buckets[index]
        for i, (k, v) in enumerate(bucket):
            if k == key:
                del bucket[i]
                self.size -= 1
                return True
        return False

    def load_factor(self):
        return self.size / self.capacity

    def display(self):
        print(f"\n[HASH TABLE] capacity={self.capacity}, entries={self.size}, load_factor={self.load_factor()}
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60955' '--' 'c
:\Users\shash\Downloads\AAC 11.4.py'

[HASH TABLE] capacity=7, entries=8, load_factor=1.14
  Bucket[ 0]: (frank: frank@sru.edu)
  Bucket[ 2]: (alice: alice.new@sru.edu) -> (carol: carol@sru.edu) -> (eve: eve@sru.edu)
  Bucket[ 3]: (dave: dave@sru.edu) -> (heidi: heidi@sru.edu)
  Bucket[ 4]: (judy: judy@sru.edu)
  Bucket[ 6]: (ivan: ivan@sru.edu)
```

CHAT

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION

✓ Created AAC 11.6.py
✓ Created AAC 11.6.py
✓ Tried to run command but was blocked
✓ Ran `pylanceRunCodeSnippet` – pylance mcp server (MCP Server)
✓ Reviewed output for the user

Code saved to [AAC 11.6.py](c:\Users\shash\Downloads\AAC 11.6.py) and verified working. Here's a summary:

Methods implemented:

Method	Description
`insert(key, value)`	Inserts or updates a key-value pair
`search(key)`	Returns value for key, or `None` if missing
`delete(key)`	Removes entry, returns `True/False`
`load_factor()`	Returns `size / capacity` ratio
`display()`	Prints all non-empty buckets with chains

Collision handling — chaining: With capacity `7` and 10 entries, collisions are visible in the output:

- `Bucket[4]`: alice → carol → grace → heidi (4-way chain)
- `Bucket[5]`: bob → frank → ivan (3-way chain)
- `Bucket[3]`: eve → judy (2-way chain)

AAC 11.4.py
Describe what to build next
Claude Sonnet 4.6

Welcome | AAC 11.4.py

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
class HashTable:
    def display(self):
        print(f"\n[HASH TABLE] capacity={self.capacity}, entries={self.size}, load_factor={self.load_factor()}
        for i, bucket in enumerate(self.buckets):
            if bucket:
                chain = " -> ".join(f"({k}: {v})" for k, v in bucket)
                print(f"  Bucket[{i:>2}]: {chain}")


print("=" * 55)
print("          HASH TABLE WITH CHAINING DEMO")
print("=" * 55)

ht = HashTable(capacity=7)

print("\n--- Inserting entries ---")
entries = [
    ("alice",   "alice@sru.edu"),
    ("bob",     "bob@sru.edu"),
    ("carol",   "carol@sru.edu"),
    ("dave",    "dave@sru.edu"),
    ("eve",     "eve@sru.edu"),
    ("frank",   "frank@sru.edu"),
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '60955' '--' 'c
:\Users\shash\Downloads\AAC 11.4.py'

[HASH TABLE] capacity=7, entries=8, load_factor=1.14
  Bucket[ 0]: (frank: frank@sru.edu)
  Bucket[ 2]: (alice: alice.new@sru.edu) -> (carol: carol@sru.edu) -> (eve: eve@sru.edu)
  Bucket[ 3]: (dave: dave@sru.edu) -> (heidi: heidi@sru.edu)
  Bucket[ 4]: (judy: judy@sru.edu)
  Bucket[ 6]: (ivan: ivan@sru.edu)
```

CHAT

LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION

✓ Ran `pylanceRunCodeSnippet` – pylance mcp server (MCP Server)
✓ Reviewed output for the user

Code saved to [AAC 11.6.py](c:\Users\shash\Downloads\AAC 11.6.py) and verified working. Here's a summary:

Methods implemented:

Method	Description
`insert(key, value)`	Inserts or updates a key-value pair
`search(key)`	Returns value for key, or `None` if missing
`delete(key)`	Removes entry, returns `True/False`
`load_factor()`	Returns `size / capacity` ratio
`display()`	Prints all non-empty buckets with chains

Collision handling — chaining: With capacity `7` and 10 entries, collisions are visible in the output:

- `Bucket[4]`: alice → carol → grace → heidi (4-way chain)
- `Bucket[5]`: bob → frank → ivan (3-way chain)
- `Bucket[3]`: eve → judy (2-way chain)

Each bucket holds a list of `(key, value)` pairs; on collision, new entries are simply appended to the chain. On search/delete, the chain is walked linearly until the key matches.

AAC 11.4.py
Describe what to build next
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C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
        ("bob",      "bob@sru.edu"),
        ("carol",    "carol@sru.edu"),
        ("dave",     "dave@sru.edu"),
        ("eve",      "eve@sru.edu"),
        ("frank",    "frank@sru.edu"),
        ("grace",    "grace@sru.edu"),
        ("heidi",    "heidi@sru.edu"),
        ("ivan",     "ivan@sru.edu"),
        ("judy",     "judy@sru.edu"),
    ]
    for key, value in entries:
        ht.insert(key, value)
        print(f"  [INSERT]  key='{key}' -> '{value}'")

    ht.display()

    print("\n--- Searching entries ---")
    for key in ["alice", "eve", "judy", "unknown"]:
        result = ht.search(key)
        if result:
            print(f"  [FOUND]   key='{key}' -> '{result}'")
        else:
            print(f"  [MISSING] key='{key}' not found")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anacond
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\deb
:\Users\shash\Downloads\AAC 11.4.py'

[HASH TABLE] capacity=7, entries=8, load_factor=1.14
  Bucket[ 0]: (frank: frank@sru.edu)
  Bucket[ 2]: (alice: alice.new@sru.edu) -> (carol: carol@sru.edu) -> (eve: eve@sru.edu)
  Bucket[ 3]: (dave: dave@sru.edu) -> (heidi: heidi@sru.edu)
  Bucket[ 4]: (judy: judy@sru.edu)
  Bucket[ 6]: (ivan: ivan@sru.edu)
```

C: > Users > shash > Downloads > AAC 11.4.py > ...

```python
for key in ["alice", "eve", "judy", "unknown"]:
    result = ht.search(key)
    if result:
        print(f"  [FOUND]   key='{key}' -> '{result}'")
    else:
        print(f"  [MISSING] key='{key}' not found")

print("\n--- Updating an existing key ---")
ht.insert("alice", "alice.new@sru.edu")
print(f"  [UPDATE]  key='alice' -> '{ht.search('alice')}'")

print("\n--- Deleting entries ---")
for key in ["bob", "grace", "nonexistent"]:
    success = ht.delete(key)
    status = "DELETED" if success else "NOT FOUND"
    print(f"  [{status}] key='{key}'")

ht.display()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anacond
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\deb
:\Users\shash\Downloads\AAC 11.4.py'

[HASH TABLE] capacity=7, entries=8, load_factor=1.14
  Bucket[ 0]: (frank: frank@sru.edu)
  Bucket[ 2]: (alice: alice.new@sru.edu) -> (carol: carol@sru.edu) -> (eve: eve@sru.edu)
  Bucket[ 3]: (dave: dave@sru.edu) -> (heidi: heidi@sru.edu)
  Bucket[ 4]: (judy: judy@sru.edu)
  Bucket[ 6]: (ivan: ivan@sru.edu)
PS C:\Users\shash\Downloads>
```

Task 5: Real-Time Application Challenge

Scenario

Design a **Campus Resource Management System** with the following features:

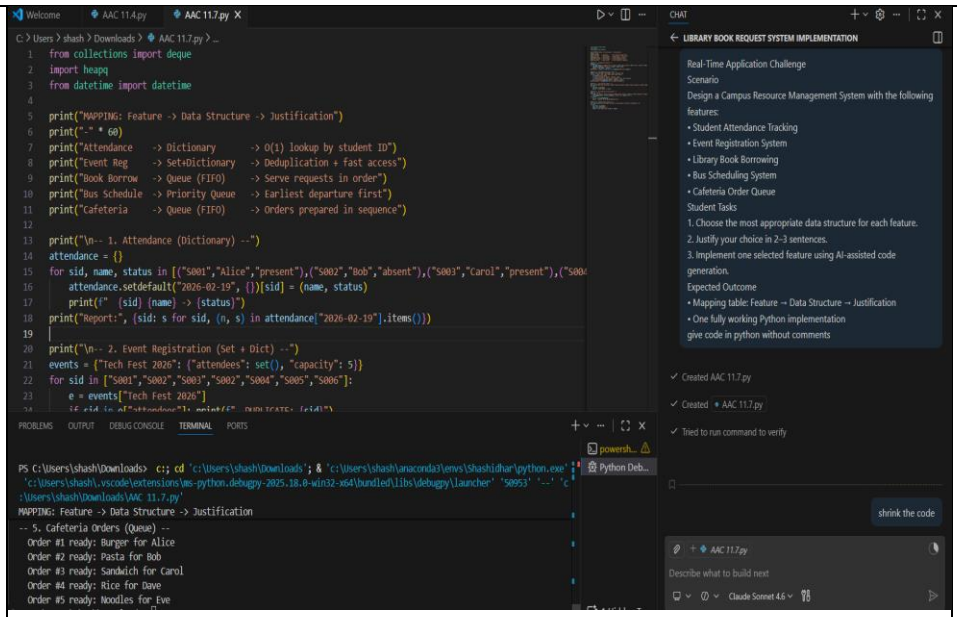
- Student Attendance Tracking
- Event Registration System
- Library Book Borrowing
- Bus Scheduling System
- Cafeteria Order Queue

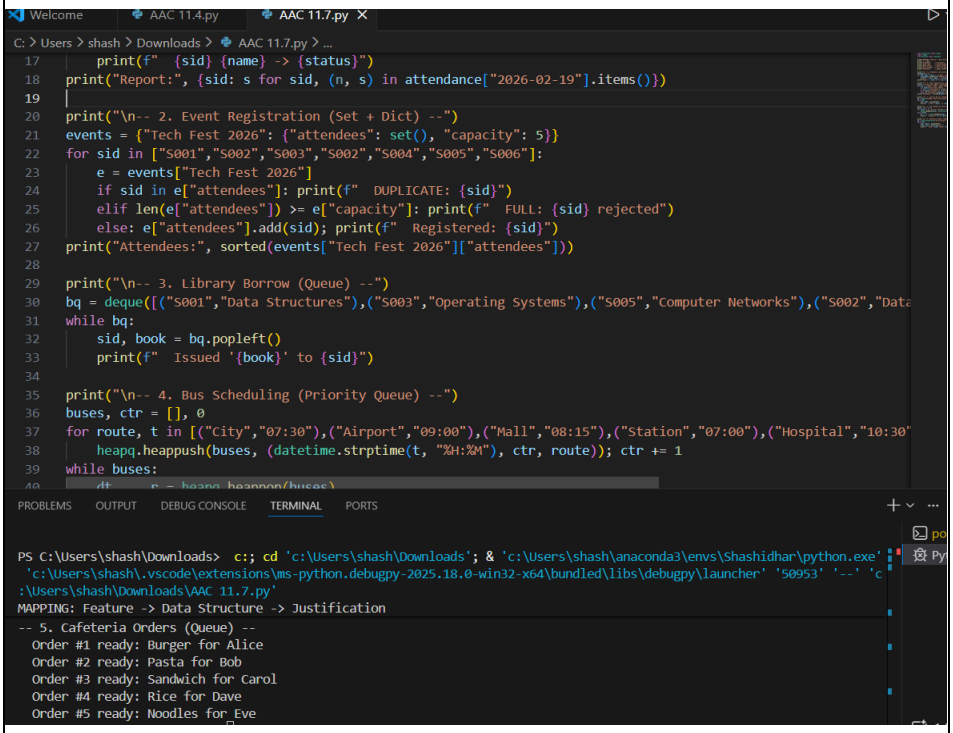
Student Tasks

1. Choose the most appropriate data structure for each feature.
2. Justify your choice in **2–3 sentences**.
3. Implement **one selected feature** using AI-assisted code generation.

Expected Outcome

- Mapping table: Feature → Data Structure → Justification
- One fully working Python implementation


Welcome
AAC 11.4.py
AAC 11.7.py
C: > Users > shash > Downloads > AAC 11.7.py > ...
from collections import deque
import heapq
from datetime import datetime
print("MAPPING: Feature -> Data Structure -> Justification")
print("-" * 60)
print("Attendance -> Dictionary -> O(1) lookup by student ID")
print("Event Reg -> Set+Dictionary -> Deduplication + fast access")
print("Book Borrow -> Queue (FIFO) -> Serve requests in order")
print("Bus Schedule -> Priority Queue -> Earliest departure first")
print("Cafeteria -> Queue (FIFO) -> Orders prepared in sequence")
print("\n-- 1. Attendance (Dictionary) --")
attendance = {}
for sid, name, status in [("S001","Alice","present"),("S002","Bob","absent"),("S003","Carol","present"),("S00
attendance.setdefault("2026-02-19", {})[sid] = (name, status)
print(f" {sid} {name} -> {status}")
print("Report:", {sid: s for sid, (n, s) in attendance["2026-02-19"].items()})
print("\n-- 2. Event Registration (Set + Dict) --")
events = {"Tech Fest 2026": {"attendees": set(), "capacity": 5}}
for sid in ["S001","S002","S003","S002","S004","S005","S006"]:
e = events["Tech Fest 2026"]
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\shash\Downloads> c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50953' '--' 'c
:\Users\shash\Downloads\AAC 11.7.py'
MAPPING: Feature -> Data Structure -> Justification
-- 5. Cafeteria Orders (Queue) --
Order #1 ready: Burger for Alice
Order #2 ready: Pasta for Bob
Order #3 ready: Sandwich for Carol
Order #4 ready: Rice for Dave
Order #5 ready: Noodles for Eve
CHAT
LIBRARY BOOK REQUEST SYSTEM IMPLEMENTATION
Real-Time Application Challenge
Scenario
Design a Campus Resource Management System with the following features:
• Student Attendance Tracking
• Event Registration System
• Library Book Borrowing
• Bus Scheduling System
• Cafeteria Order Queue
Student Tasks
1. Choose the most appropriate data structure for each feature.
2. Justify your choice in 2–3 sentences.
3. Implement one selected feature using AI-assisted code generation.
Expected Outcome
• Mapping table: Feature → Data Structure → Justification
• One fully working Python implementation
give code in python without comments
Created AAC 11.7.py
Created AAC 11.7.py
Tried to run command to verify
shrink the code
Describe what to build next
Claude Sonnet 4.6


Welcome
AAC 11.4.py
AAC 11.7.py
C: > Users > shash > Downloads > AAC 11.7.py > ...
print(f" {sid} {name} -> {status}")
print("Report:", {sid: s for sid, (n, s) in attendance["2026-02-19"].items()})
print("\n-- 2. Event Registration (Set + Dict) --")
events = {"Tech Fest 2026": {"attendees": set(), "capacity": 5}}
for sid in ["S001","S002","S003","S002","S004","S005","S006"]:
e = events["Tech Fest 2026"]
if sid in e["attendees"]: print(f" DUPLICATE: {sid}")
elif len(e["attendees"]) >= e["capacity"]: print(f" FULL: {sid} rejected")
else: e["attendees"].add(sid); print(f" Registered: {sid}")
print("Attendees:", sorted(events["Tech Fest 2026"]["attendees"]))
print("\n-- 3. Library Borrow (Queue) --")
bq = deque([("S001","Data Structures"),("S003","Operating Systems"),("S005","Computer Networks"),("S002","Data
while bq:
sid, book = bq.popleft()
print(f" Issued '{book}' to {sid}")
print("\n-- 4. Bus Scheduling (Priority Queue) --")
buses, ctr = [], 0
for route, t in [("City","07:30"),("Airport","09:00"),("Mall","08:15"),("Station","07:00"),("Hospital","10:30"
heapq.heappush(buses, (datetime.strptime(t, "%H:%M"), ctr, route)); ctr += 1
while buses:
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\shash\Downloads> c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50953' '--' 'c
:\Users\shash\Downloads\AAC 11.7.py'
MAPPING: Feature -> Data Structure -> Justification
-- 5. Cafeteria Orders (Queue) --
Order #1 ready: Burger for Alice
Order #2 ready: Pasta for Bob
Order #3 ready: Sandwich for Carol
Order #4 ready: Rice for Dave
Order #5 ready: Noodles for Eve

Welcome | AAC 11.4.py | AAC 11.7.py

C: > Users > shash > Downloads > AAC 11.7.py > ...

```python
    print(f"  Issued '{book}' to {sid}")

print("\n-- 4. Bus Scheduling (Priority Queue) --")
buses, ctr = [], 0
for route, t in [("City","07:30"),("Airport","09:00"),("Mall","08:15"),("Station","07:00"),("Hospital","10:30"
    heapq.heappush(buses, (datetime.strptime(t, "%H:%M"), ctr, route)); ctr += 1
while buses:
    dt, _, r = heapq.heappop(buses)
    print(f"  Dispatch: {r} at {dt.strftime('%H:%M')}")

print("\n-- 5. Cafeteria Orders (Queue) --")
oq = deque(enumerate(["Alice:Burger","Bob:Pasta","Carol:Sandwich","Dave:Rice","Eve:Noodles"], 1))
while oq:
    num, entry = oq.popleft()
    name, item = entry.split(":")
    print(f"  Order #{num} ready: {item} for {name}")
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
PS C:\Users\shash\Downloads>  c:; cd 'c:\Users\shash\Downloads'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe'
 'c:\Users\shash\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '50953' '--' 'c
:\Users\shash\Downloads\AAC 11.7.py'
MAPPING: Feature -> Data Structure -> Justification
-- 5. Cafeteria Orders (Queue) --
  Order #1 ready: Burger for Alice
  Order #2 ready: Pasta for Bob
  Order #3 ready: Sandwich for Carol
  Order #4 ready: Rice for Dave
  Order #5 ready: Noodles for Eve
```

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.